

The topic is

- A topic is a subject or a theme that is discussed or written about.
- A topic can be broad or narrow, depending on the scope and purpose of the discussion or writing.
- A topic can be chosen by the writer or speaker, or assigned by a teacher or an editor.
- A topic can be expressed as a word, a phrase, a question, or a statement.
- A topic can be related to various fields of study, such as science, history, literature, art, etc.
- A topic can be informative, persuasive, argumentative, descriptive, narrative, or analytical, depending on the type and goal of the communication.
- A topic can be supported by evidence, examples, facts, opinions, or arguments, depending on the genre and audience of the communication.
- A topic can be organized in different ways, such as chronological, spatial, cause and effect, problem and solution, compare and contrast, etc., depending on the structure and logic of the communication.

FUNDAMENTALS OF ELECTRICAL ENGINEERING

- Electrical engineering is the branch of engineering that deals with the study, design, and application of electrical and electronic systems, devices, and circuits.
- Electrical engineering covers a wide range of topics, such as power generation and distribution, communication and information systems, control and automation, signal processing and analysis, and computer hardware and software.
- Some of the fundamental concepts of electrical engineering are:
 - Voltage, current, and power: Voltage is the difference in electric potential between two points, current is the rate of flow of electric charge, and power is the rate of doing work or transferring energy.
 - Ohm's law: Ohm's law states that the current through a conductor is directly proportional to the voltage across it and inversely proportional to its resistance.
 - Kirchhoff's laws: Kirchhoff's laws are two rules that govern the conservation of charge and energy in electrical circuits. The first law states that the algebraic sum of currents at any node is zero. The second law states that the algebraic sum of voltages around any loop is zero.
 - Series and parallel circuits: Series circuits are circuits in which the components are connected end to end, so that the same current flows through all of them. Parallel circuits are circuits in which the compo-

nents are connected across common points, so that the voltage across each component is the same.

- Equivalent circuits: Equivalent circuits are simplified models of complex circuits that have the same input-output characteristics as the original circuit. For example, a series combination of resistors can be replaced by a single resistor with the same total resistance.
- AC and DC circuits: AC circuits are circuits that operate with alternating current, which is a current that changes direction periodically. DC circuits are circuits that operate with direct current, which is a current that flows in one direction only. AC circuits are more common in power systems, while DC circuits are more common in electronic devices.
- Transformers: Transformers are devices that use electromagnetic induction to transfer electrical energy from one circuit to another, usually with a change in voltage and current. Transformers consist of two coils of wire, called the primary and secondary, that are wound around a magnetic core. When an AC voltage is applied to the primary coil, it induces an AC voltage in the secondary coil, according to the ratio of the number of turns in each coil.
- Three-phase circuits: Three-phase circuits are circuits that use three AC voltages that are 120 degrees out of phase with each other. Three-phase circuits are more efficient and reliable than single-phase circuits for transmitting and distributing power over long distances.
- Diodes: Diodes are semiconductor devices that allow current to flow in one direction only. Diodes are used for rectification, which is the process of converting AC to DC, and for switching, which is the process of controlling the flow of current in a circuit.
- Transistors: Transistors are semiconductor devices that can amplify or switch electrical signals. Transistors consist of three terminals, called the base, collector, and emitter, that are made of different types of semiconductor materials. By applying a small voltage to the base, the transistor can control the current flow between the collector and the emitter. Transistors are the building blocks of modern electronic devices, such as computers and smartphones.

Unit 1 - DC Circuits

- A DC circuit is a circuit that carries direct current (DC), which is a type of electric current that flows in one direction only.
- The main components of a DC circuit are a voltage source, such as a battery or a generator, and a load, such as a resistor, a capacitor, an inductor, or a combination of them.
- The voltage source provides the electric potential difference that drives the current through the circuit, while the load consumes the electric power and converts it into other forms of energy, such as heat, light, or motion.
- The basic laws that govern the behavior of DC circuits are Ohm's law,

Kirchhoff's current law, and Kirchhoff's voltage law.

- Ohm's law states that the current (I) through a resistor (R) is proportional to the voltage (V) across it, or $I = V/R$.
- Kirchhoff's current law states that the algebraic sum of the currents entering and leaving any node in a circuit is zero, or $I = 0$.
- Kirchhoff's voltage law states that the algebraic sum of the voltages around any closed loop in a circuit is zero, or $V = 0$.
- These laws can be used to analyze and solve DC circuits by applying various methods, such as nodal analysis, mesh analysis, superposition, Thevenin's theorem, Norton's theorem, and maximum power transfer.

Electrical circuit elements (R , L and C)

- Electrical circuit elements are components that can store, dissipate, or transfer electrical energy in a circuit.
- The most common circuit elements are resistors (R), inductors (L), and capacitors (C).
- Resistors are devices that oppose the flow of electric current and convert electrical energy into heat. The resistance (R) of a resistor is measured in ohms (Ω) and depends on the material, shape, and temperature of the resistor. The voltage (V) across a resistor is proportional to the current (I) through it, according to Ohm's law: $V = IR$.
- Inductors are devices that store electrical energy in a magnetic field when current flows through them. The inductance (L) of an inductor is measured in henries (H) and depends on the number of turns, the shape, and the core material of the coil. The voltage across an inductor is proportional to the rate of change of current through it, according to Faraday's law: $V = L(dI/dt)$.
- Capacitors are devices that store electrical energy in an electric field when a voltage is applied across them. The capacitance (C) of a capacitor is measured in farads (F) and depends on the area, distance, and dielectric material of the plates. The voltage across a capacitor is proportional to the charge (Q) stored on it, according to the definition of capacitance: $V = Q/C$.

Concept of active and passive elements

- Active and passive elements are two types of electronic circuit elements that differ in their ability to supply or consume energy.
- Active elements are those that can produce electrical energy or provide amplification to a circuit. They can control the direction and magnitude of current or voltage in a circuit. They have the function of gain for electronic circuits. Examples of active elements are batteries, generators, transistors, op-amps, etc.
- Passive elements are those that can only receive, store, or dissipate

electrical energy. They cannot control the current or voltage in a circuit on their own. They do not have the function of gain for electronic circuits. Examples of passive elements are resistors, capacitors, inductors, diodes, etc.

- Active and passive elements are used for different purposes in electrical circuits. Active elements are used for current and voltage control, signal processing, amplification, oscillation, etc. Passive elements are used for energy storage, discharge, filtering, impedance matching, etc.
- The following table summarizes the main differences between active and passive elements:

Comparison between Active and Passive	Active Element	Passive Element
Produces energy to another circuit in the form of current or voltage	Yes	No
Impervious in nature i.e. cannot supply energy to the circuit	No	Yes
Controls current direction	Yes	No
Have function of gain for electronic circuits	Yes	No
Used for current control and voltage control, signal processing, amplification, oscillation, etc.	Yes	No
Used for energy storage, discharge, filtering, impedance matching, etc.	No	Yes

Voltage and Current Sources

- A voltage source is a device that provides a constant voltage across its terminals, regardless of the current drawn from it . An example of a voltage source is a battery.
- A current source is a device that provides a constant current through its terminals, regardless of the voltage across it . An example of a current source is a solar cell.
- Both voltage and current sources are idealized models that do not exist in reality. In practice, every source has some internal resistance that limits its ability to maintain a constant voltage or current.
- The internal resistance of a voltage source is the resistance in series with the ideal voltage source. The internal resistance of a current source is the resistance in parallel with the ideal current source.
- The voltage and current sources can be classified into two types: independent and dependent. Independent sources are those that do not depend on any other quantity in the circuit. Dependent sources are those that

depend on some other voltage or current in the circuit .

- The symbols for independent and dependent voltage and current sources are shown below:

Symbol	Type	Description
Independent voltage source	Independent voltage source	Provides a constant voltage across its terminals
Dependent voltage source	Dependent voltage source	Provides a voltage that is proportional to some other voltage or current in the circuit
Independent current source	Independent current source	Provides a constant current through its terminals
Dependent current source	Dependent current source	Provides a current that is proportional to some other voltage or current in the circuit

- The sign convention for passive components and sources is that the current enters the positive terminal and leaves the negative terminal of a voltage source, and the voltage is positive across a current source in the direction of the current. This is shown below:

Symbol	Sign convention
Sign convention for voltage source	Sign convention for voltage source
Sign convention for current source	Sign convention for current source

- The voltage and current sources are the basic elements that supply energy to the electrical circuits. The voltage and current sources can be used to analyze the behavior of the circuits using various methods such as Kirchhoff's laws, mesh analysis, nodal analysis, etc.

Concept of linearity

- Linearity is a property of an element or a system that describes a linear relationship between cause and effect.
- A linear element or system obeys the principles of homogeneity and additivity.
- Homogeneity means that if the input is scaled by a factor, the output is also scaled by the same factor.
- Additivity means that if the input is the sum of two or more signals, the output is also the sum of the corresponding outputs.

- In electrical engineering, linearity is often applied to the current-voltage characteristics of a component or a circuit.
- A linear component or circuit has a current-voltage curve that is a straight line, meaning that the current is proportional to the voltage.
- Examples of linear components are resistors, capacitors, and inductors.
- A linear circuit is a circuit that consists of only linear components and independent sources.
- A linear system is a system that can be modeled by a linear equation or a set of linear equations.
- Examples of linear systems are amplifiers, filters, and feedback controllers.
- Linearity is a useful property because it simplifies the analysis and design of electrical systems.
- One of the benefits of linearity is the principle of superposition, which states that the response of a linear system to a combination of inputs is the same as the sum of the responses to each input separately.
- Superposition can help reduce the complexity of solving a circuit with multiple sources.

Unilateral and Bilateral Elements

- Unilateral and bilateral elements are the two different types of electrical / electronic circuit elements on the basis of their V-I characteristics on reversal of voltage polarity.
- Unilateral elements are those that allow the current in only one direction and offer different impedance in different directions of current flow. Examples: Diodes, Transistors .
- Bilateral elements are those that allow the current in both directions and offer the same impedance in either direction of current flow. Examples: Resistors, Inductors, Capacitors .
- Unilateral and bilateral circuits are the circuits that contain only unilateral or only bilateral elements, or a combination of both.
- Unilateral and bilateral circuits can be used for various applications such as amplifiers, oscillators, power supplies and more.

Kirchhoff's laws for the notes of the Unit 1 - DC Circuits in the subject of FUNDAMENTALS OF ELECTRICAL ENGINEERING

- Kirchhoff's laws are two rules that relate the currents and voltages in a circuit. They are based on the conservation of charge and energy principles.
- Kirchhoff's current law (KCL) states that the algebraic sum of the currents entering a node (or a closed boundary) is zero. This means that the total charge entering a node is equal to the total charge leaving a node.
- Kirchhoff's voltage law (KVL) states that the algebraic sum of the voltages around a loop (or a closed path) is zero. This means that the total energy gained by the charges in a loop is equal to the total energy lost by the

charges in the same loop.

- Kirchhoff's laws can be used to analyze any circuit, whether it is linear or nonlinear, passive or active, lumped or distributed, etc.
- Kirchhoff's laws can be applied to any node or loop in a circuit, as long as they are independent. A node is independent if it is not contained within another node. A loop is independent if it does not enclose another loop.
- Kirchhoff's laws can be written in matrix form, using the node-voltage method or the mesh-current method. These methods can simplify the solution of large and complex circuits.

Mesh and Nodal Methods of Analysis

- Mesh and nodal methods are two systematic techniques for analyzing electrical circuits by applying Kirchhoff's laws.
- Mesh analysis is based on Kirchhoff's voltage law (KVL), which states that the sum of voltages around any closed loop in a circuit is zero.
- Nodal analysis is based on Kirchhoff's current law (KCL), which states that the sum of currents entering and leaving any node in a circuit is zero.
- Both methods can be used to solve for unknown currents and voltages in a circuit, but they have different advantages and disadvantages.

Mesh Analysis

- A mesh is a loop in a circuit that does not contain any other loops within it.
- Mesh analysis involves assigning a current variable to each mesh and writing a KVL equation for each mesh.
- The number of equations and unknowns is equal to the number of meshes in the circuit.
- Mesh analysis is more suitable for circuits with many loops and few nodes.
- Mesh analysis can be simplified by using the supermesh technique, which combines two adjacent meshes that share a current source.

Nodal Analysis

- A node is a point in a circuit where two or more elements are connected.
- Nodal analysis involves assigning a voltage variable to each node and writing a KCL equation for each node except the reference node, which is usually chosen as the ground node with zero voltage.
- The number of equations and unknowns is equal to the number of nodes minus one in the circuit.
- Nodal analysis is more suitable for circuits with many nodes and few loops.
- Nodal analysis can be simplified by using the supernode technique, which combines two adjacent nodes that share a voltage source.

Unit 2 - Steady State Analysis of Single Phase AC Circuits

- Single phase AC circuits are electrical circuits that are powered by alternating current (AC) sources with a single frequency and phase.
- AC sources produce sinusoidal voltages and currents that vary periodically in time and direction.
- The frequency of an AC source is the number of cycles per second, measured in hertz (Hz).
- The phase of an AC source is the angle of the sinusoidal waveform relative to a reference point, measured in degrees or radians.
- The peak value of an AC voltage or current is the maximum value attained by the sinusoid in either direction.
- The root mean square (RMS) value of an AC voltage or current is the effective value that produces the same power as a constant DC voltage or current of the same magnitude.
- The RMS value of a sinusoid is equal to its peak value divided by the square root of 2.
- The average value of an AC voltage or current over one cycle is zero, since the positive and negative halves cancel out.
- The average power delivered by an AC source to a resistive load is equal to the product of the RMS voltage, the RMS current, and the power factor.
- The power factor of an AC circuit is the ratio of the average power to the apparent power, which is the product of the RMS voltage and the RMS current.
- The power factor is also equal to the cosine of the phase angle between the voltage and the current.
- The power factor can range from 0 to 1, where 0 means the voltage and current are out of phase by 90 degrees, and 1 means they are in phase.
- The power factor can be improved by adding capacitors or inductors to the circuit, which can cancel out the reactive power and reduce the phase angle.
- The impedance of an AC circuit is the ratio of the RMS voltage to the RMS current, measured in ohms.
- The impedance is a complex number that has a real part (resistance) and an imaginary part (reactance).
- The reactance of an AC circuit is the opposition to the change of voltage or current, caused by capacitors or inductors.
- The reactance of a capacitor is inversely proportional to the frequency and the capacitance, and has a negative sign.
- The reactance of an inductor is directly proportional to the frequency and the inductance, and has a positive sign.
- The impedance of a series AC circuit is the sum of the impedances of the individual components.
- The impedance of a parallel AC circuit is the reciprocal of the sum of the reciprocals of the impedances of the individual components.
- The phasor diagram of an AC circuit is a graphical representation of the

voltages and currents as vectors, with their magnitudes and phases.

- The phasor diagram can be used to find the impedance, the power factor, and the power of an AC circuit, using trigonometry and vector algebra.
- The Kirchhoff's laws of an AC circuit are the same as those of a DC circuit, except that the voltages and currents are complex numbers and must be added or subtracted using phasor notation.
- The mesh analysis of an AC circuit is a method of finding the currents in the loops of the circuit, by applying Kirchhoff's voltage law and solving a system of linear equations.
- The nodal analysis of an AC circuit is a method of finding the voltages at the nodes of the circuit, by applying Kirchhoff's current law and solving a system of linear equations.
- The superposition theorem of an AC circuit states that the response of a linear circuit to multiple AC sources is equal to the algebraic sum of the responses to each source acting alone, with all other sources replaced by their internal impedances.
- The Thevenin's theorem of an AC circuit states that any linear circuit with multiple AC sources can be replaced by an equivalent circuit with a single AC source and an impedance, where the source voltage is the open circuit voltage and the impedance is the Thevenin's impedance.
- The Norton's theorem of an AC circuit states that any linear circuit with multiple AC sources can be replaced by an equivalent circuit with a single AC current source and an impedance, where the source current is the short circuit current and the impedance is the Norton's impedance.
- The maximum power transfer theorem of an AC circuit states that the maximum power is delivered to a load when the load impedance is equal to the complex conjugate of the source impedance.

Representation of Sinusoidal waveforms – Average and effective values

- A sinusoidal waveform is a periodic function that oscillates between positive and negative peak values with a constant frequency.
- The peak value of a sinusoidal waveform is the maximum or minimum amplitude that it reaches in one cycle.
- The average value of a sinusoidal waveform is the arithmetic mean of the positive and negative half cycles over one period.
- The average value of a sinusoidal waveform is zero if it is symmetrical about the x-axis, and nonzero if it is shifted or offset.
- The average value of a sinusoidal waveform can be calculated by multiplying the peak value by 0.637, which is a constant derived from the integration of the sine function over one period.
- The effective value or root mean square (RMS) value of a sinusoidal waveform is the square root of the average of the squares of the instantaneous values over one period.
- The RMS value of a sinusoidal waveform is a measure of the power or heating effect that it can produce in a resistive load.

- The RMS value of a sinusoidal waveform is equal to the peak value divided by the square root of 2, or multiplied by 0.707, which is another constant derived from the integration of the sine function over one period.
- The RMS value of a sinusoidal waveform is independent of the phase angle or the position of the waveform with respect to the reference axis.
- The form factor of a sinusoidal waveform is the ratio of the RMS value to the average value. For a pure sine wave, the form factor is 1.11.
- The form factor of a sinusoidal waveform indicates how closely it resembles a pure sine wave. The higher the form factor, the more distorted the waveform is.
- The peak factor of a sinusoidal waveform is the ratio of the peak value to the RMS value. For a pure sine wave, the peak factor is 1.414.
- The peak factor of a sinusoidal waveform indicates how sharp the peaks are. The higher the peak factor, the more peaked the waveform is.
- A phasor is a vector representation of a sinusoidal waveform that rotates counterclockwise with an angular frequency equal to the frequency of the waveform.
- A phasor has a magnitude equal to the peak value of the waveform and a phase angle equal to the phase shift of the waveform with respect to a reference axis.
- A phasor can be used to simplify the analysis of AC circuits by converting time-domain equations into phasor-domain equations using complex numbers.
- A resistor is a passive component that opposes the flow of electric current and dissipates energy as heat.
- A resistor has a constant resistance that does not depend on the frequency or phase of the applied voltage or current.
- A resistor obeys Ohm's law, which states that the voltage across a resistor is proportional to the current through it, and the constant of proportionality is the resistance.
- A resistor in an AC circuit has the same voltage and current waveforms, which are in phase with each other.

Form and peak factors for the notes of the Unit 2 - Steady State Analysis of Single Phase AC Circuits in the subject of FUNDAMENTALS OF ELECTRICAL ENGINEERING

- An alternating current (AC) waveform is a periodic function that varies in magnitude and direction with time. The waveform can have different shapes, such as sinusoidal, triangular, square, etc.
- The peak value of an AC waveform is the maximum absolute value that it attains in one cycle. The peak value is denoted by V_p for voltage and I_p for current.
- The root mean square (RMS) value of an AC waveform is the effective value that produces the same heating effect as a direct current (DC) of the same magnitude. The RMS value is denoted by V_{rms} for voltage and

Irms for current. The RMS value can be calculated by taking the square root of the average of the squares of the instantaneous values over one cycle.

- The average value of an AC waveform is the arithmetic mean of the instantaneous values over one cycle. The average value is denoted by V_{avg} for voltage and I_{avg} for current. The average value can be calculated by integrating the instantaneous values over one cycle and dividing by the period.
- The form factor of an AC waveform is the ratio of the RMS value to the average value. The form factor is denoted by F_f for voltage and current. The form factor can be used to describe the shape of the AC waveform. The form factor for a pure sinusoidal waveform is 1.11, while the form factor for a square waveform is 1.
- The peak factor of an AC waveform is the ratio of the peak value to the RMS value. The peak factor is denoted by F_p for voltage and current. The peak factor can be used to measure the peakiness or crest of the AC waveform. The peak factor for a pure sinusoidal waveform is 1.414, while the peak factor for a square waveform is 1.

Analysis of single phase AC Circuits consisting R-L-C combination (Series and Parallel)

- A single phase AC circuit is a circuit that is powered by a single alternating voltage or current source.
- A R-L-C circuit is a circuit that contains a resistor (R), an inductor (L) and a capacitor (C) as its elements.
- A R-L-C circuit can be connected in series or parallel, depending on how the elements are arranged with respect to the voltage or current source.
- In a series R-L-C circuit, the elements are connected end-to-end, so that the same current flows through all of them. The total impedance (Z) of the circuit is the sum of the individual impedances of the elements.
- In a parallel R-L-C circuit, the elements are connected across the voltage source, so that the same voltage is applied to all of them. The total admittance (Y) of the circuit is the sum of the individual admittances of the elements.
- The impedance (Z) and the admittance (Y) are complex quantities that have both magnitude and phase angle. They depend on the frequency (f) of the AC source, as well as the values of R, L and C.
- The impedance (Z) and the admittance (Y) can be represented by phasor diagrams or complex numbers in rectangular or polar forms. They can also be related by the equation $Y = 1/Z$.
- The impedance (Z) and the admittance (Y) can be decomposed into two components: resistance (R) or conductance (G), which are in phase with the voltage or current; and reactance (X) or susceptance (B), which are 90 degrees out of phase with the voltage or current.
- The reactance (X) and the susceptance (B) can be further divided into

two types: inductive (XL or BL), which are positive and lag the voltage or current by 90 degrees; and capacitive (XC or BC), which are negative and lead the voltage or current by 90 degrees.

- The impedance (Z) and the admittance (Y) can be represented by triangles, where the hypotenuse is Z or Y, the horizontal side is R or G, and the vertical side is X or B. The angle between Z or Y and R or G is called the phase angle (ϕ), which indicates the phase difference between the voltage and the current in the circuit.
- The power (P) delivered by the AC source to the R-L-C circuit can be calculated by the equation $P = VI \cos \phi$, where V and I are the rms values of the voltage and the current, and $\cos \phi$ is the power factor. The power factor is a measure of how efficiently the circuit uses the power from the source. It ranges from 0 to 1, where 1 means that the circuit is purely resistive and all the power is dissipated as heat, and 0 means that the circuit is purely reactive and no power is transferred to the circuit.
- The power (P) can be decomposed into two components: active power (Pa), which is the power dissipated as heat by the resistor; and reactive power (Pr), which is the power stored and released by the inductor and the capacitor. The active power can be calculated by $P_a = I^2 R$, and the reactive power can be calculated by $P_r = I^2 X$.
- The power (P) can also be expressed by the equation $P = S \cos \phi$, where S is the apparent power, which is the product of the rms values of the voltage and the current. The apparent power can be calculated by $S = VI$, and it represents the maximum power that the source can deliver to the circuit. The apparent power can be decomposed into two components: active power (Pa) and reactive power (Pr). The apparent power can be represented by a right-angled triangle, where the hypotenuse is S, the horizontal side is Pa, and the vertical side is Pr. The angle between S and Pa is the same as the phase angle (ϕ).
- The analysis of series and parallel R-L-C circuits involves finding the values of the impedance (Z) or the admittance (Y), the phase angle (ϕ), the power factor ($\cos \phi$), the active

Apparent, active and reactive power

- Apparent power is the product of the RMS voltage and current in an AC circuit, without reference to the phase angle between them. It is measured in volt-amperes (VA) and symbolized by S. It represents the total power supplied to the circuit, but not all of it may be useful for doing work.
- Active power is the component of the apparent power that is in phase with the voltage and does work in the circuit. It is measured in watts (W) and symbolized by P. It represents the average power consumed by the resistive elements in the circuit, such as heaters, lamps, and motors.
- Reactive power is the component of the apparent power that is out of phase with the voltage and does not do work in the circuit. It is measured in volt-amperes reactive (VAR) and symbolized by Q. It represents the

power stored and released by the reactive elements in the circuit, such as capacitors and inductors.

- The relationship between apparent, active and reactive power can be expressed by the power triangle, where S is the hypotenuse, P is the adjacent side, and Q is the opposite side. The angle between S and P is the power factor angle, which indicates how much of the apparent power is useful for doing work.
- The power factor is the ratio of the active power to the apparent power, or the cosine of the power factor angle. It is a dimensionless number between 0 and 1, where 1 means that all the apparent power is active power, and 0 means that all the apparent power is reactive power. A high power factor indicates a more efficient use of power, while a low power factor indicates a more wasteful use of power.

Power factor

- Power factor is a dimensionless quantity that measures how effectively the electric power is converted into useful work in an AC circuit.
- Power factor is defined as the ratio of real power (P) to apparent power (S) in an AC circuit. Real power is the power that performs useful work, such as turning a motor or lighting a bulb. Apparent power is the product of the voltage and current in the circuit, regardless of the phase angle between them.
- Power factor can be expressed as: $PF = P / S$, where P is the real power in watts (W) or kilowatts (kW), and S is the apparent power in volt-amperes (VA) or kilovolt-amperes (kVA).
- Power factor can also be calculated as the cosine of the phase angle (θ) between the voltage and current waveforms in an AC circuit. If the current is in phase with the voltage, the power factor is 1, which means that all the apparent power is converted into real power. If the current is out of phase with the voltage, the power factor is less than 1, which means that some of the apparent power is wasted as reactive power. Reactive power is the power that oscillates between the source and the load, and does not perform any useful work.
- Power factor can be expressed as: $PF = \cos \theta$, where θ is the phase angle between the voltage and current waveforms.
- Power factor can range from 0 to 1, depending on the type and configuration of the load in the circuit. A purely resistive load, such as a heater or a lamp, has a power factor of 1, because the current and voltage are in phase. A purely inductive or capacitive load, such as a motor or a capacitor, has a power factor of 0, because the current and voltage are 90 degrees out of phase. A mixed load, such as a fluorescent light or a computer, has a power factor between 0 and 1, because the current and

voltage are not in phase but not 90 degrees apart either.

- Power factor is important for the efficiency and reliability of the power system. A low power factor means that more current is required to deliver the same amount of real power, which increases the losses and heating in the transmission lines and transformers. A low power factor also reduces the capacity of the power system to supply more loads. A high power factor means that less current is required to deliver the same amount of real power, which reduces the losses and heating in the transmission lines and transformers. A high power factor also increases the capacity of the power system to supply more loads.
- Power factor can be improved by using power factor correction devices, such as capacitors, inductors, or synchronous motors, that can adjust the phase angle between the voltage and current in the circuit. Power factor correction can reduce the apparent power and increase the real power, which improves the efficiency and reliability of the power system.

Concept of Resonance in Series and Parallel Circuits

- Resonance is a condition that occurs in AC circuits when the reactive power of the capacitors and inductors become equal.
- Resonance can happen in circuits where capacitors and inductors are connected in series or in parallel.
- In series resonance, the circuit has minimum impedance and maximum current at the resonant frequency. The series resonance circuit is also known as the acceptor circuit because it accepts the maximum power from the source at the resonant frequency.
- In parallel resonance, the circuit has maximum impedance and minimum current at the resonant frequency. The parallel resonance circuit is also known as the rejector circuit because it rejects the maximum power from the source at the resonant frequency.
- The resonant frequency of a series or parallel circuit is given by the formula: $f_0 = 1 / (2 \sqrt{LC})$, where L is the inductance and C is the capacitance.
- The quality factor or Q factor of a series or parallel circuit is a measure of how sharp the resonance is. It is defined as the ratio of the reactive power to the resistive power at the resonant frequency.
- The Q factor of a series circuit is given by the formula: $Q_{\text{series}} = X_L / R$, where X_L is the inductive reactance and R is the resistance.
- The Q factor of a parallel circuit is given by the formula: $Q_{\text{parallel}} = R_T / X_L$, where R_T is the total parallel resistance and X_L is the inductive reactance.
- The Q factor of a series or parallel circuit determines the bandwidth or the range of frequencies around the resonant frequency where the circuit behaves as a resonant circuit.
- The bandwidth of a series or parallel circuit is given by the formula: $B =$

f_0 / Q , where f_0 is the resonant frequency and Q is the quality factor.

- The higher the Q factor, the narrower the bandwidth and the sharper the resonance. The lower the Q factor, the wider the bandwidth and the flatter the resonance.
- The presence of resistance in a series or parallel circuit can affect the resonant frequency and the Q factor. Resistance can cause the resonant frequency to shift from the calculated value and the Q factor to decrease.
- The effect of resistance on the resonant frequency and the Q factor is called antiresonance. Antiresonance can be minimized by using low-resistance components and wires in the circuit.

Bandwidth and Quality Factor for the Notes of the Unit 2 - Steady State Analysis of Single Phase AC Circuits in the Subject of Fundamentals of Electrical Engineering

- Bandwidth of a resonant circuit is the range of frequencies over which the current or voltage amplitude is equal to or greater than 70.7% of its maximum value.
- Quality factor of a resonant circuit is a dimensionless quantity that measures how underdamped the circuit is, or how narrow the bandwidth is in relation to the resonant frequency.
- The quality factor of an AC circuit is related to bandwidth as inversely proportional, i.e., $Q = f_r / BW$, where f_r is the resonant frequency and BW is the bandwidth.
- The quality factor and the bandwidth of a resonant circuit depend on whether the circuit is series or parallel. For a series circuit, $Q = X_L / R$ and $BW = R / L$, where X_L is the inductive reactance, R is the resistance, and L is the inductance. For a parallel circuit, $Q = R / X_L$ and $BW = L / RC$, where C is the capacitance.
- A high Q resonant circuit has a narrow bandwidth and a sharp resonance peak, whereas a low Q resonant circuit has a wide bandwidth and a flat resonance peak.

Three phase balanced circuits

- A three phase balanced circuit is a type of alternating current circuit that consists of three voltage sources and three loads connected in either star or delta configuration.
- The three voltage sources have equal magnitudes and frequencies, but are phase shifted by 120 degrees with respect to each other. The three loads have equal impedances and power factors.
- In a three phase balanced circuit, the total power delivered by the sources is equal to the total power consumed by the loads, and the neutral current is zero.
- The advantages of using three phase balanced circuits are higher efficiency, lower cost, and smoother operation of electrical machines and devices.

- The analysis of a three phase balanced circuit can be simplified by using the following steps:
 - Choose a reference phase, usually phase a, and assign the positive direction of the current and voltage in that phase.
 - Draw the phasor diagram of the voltages and currents in the reference phase, using the phase sequence abc or acb depending on the polarity of the sources.
 - Apply Kirchhoff's voltage law and Kirchhoff's current law to the reference phase, and solve for the unknown quantities.
 - Use the phase shift of 120 degrees to find the voltages and currents in the other two phases, b and c.
 - Calculate the power and power factor of each phase and the total power and power factor of the circuit.

Voltage and Current Relations in Star and Delta Connections

- A three-phase system can be connected in two different styles: star or delta .
- In a star connection, the three phase windings are connected in such a way that one end of each winding is joined together to form a common neutral point, and the other end is connected to the line terminals .
- In a delta connection, the three phase windings are connected in a closed loop, such that each winding is in series with the other two, and the line terminals are connected at the junctions of the windings .
- The voltage and current relations in star and delta connections are as follows :
 - In a star connection, the line voltage (the voltage between any two line terminals) is equal to the square root of three times the phase voltage (the voltage across any phase winding). Mathematically, $V_L = \sqrt{3} V_P$.
 - In a star connection, the line current (the current in any line) is equal to the phase current (the current in the corresponding phase winding). Mathematically, $I_L = I_P$.
 - In a delta connection, the line voltage is equal to the phase voltage. Mathematically, $V_L = V_P$.
 - In a delta connection, the line current is equal to the square root of three times the phase current. Mathematically, $I_L = \sqrt{3} I_P$.
- The power transmitted by both the connections is the same, as long as the phase voltage and phase current are the same in both cases. Mathematically, $P = 3 V_P I_P \cos \phi$, where ϕ is the power factor angle.

Unit 3 - Transformers

- A transformer is a device that converts alternating current (AC) from one voltage level to another voltage level by using the principle of electromagnetic induction .
- A transformer consists of two or more coils of wire, called the primary and secondary coils, that are wound around a common core, usually made of iron or steel .
- The primary coil is connected to the AC source, and the secondary coil is connected to the load. When an AC voltage is applied to the primary coil, it creates a changing magnetic flux in the core, which induces an AC voltage in the secondary coil .
- The ratio of the number of turns of wire in the primary and secondary coils determines the voltage transformation. If the secondary coil has more turns than the primary coil, the output voltage is higher than the input voltage, and the transformer is called a step-up transformer. If the secondary coil has fewer turns than the primary coil, the output voltage is lower than the input voltage, and the transformer is called a step-down transformer .
- Transformers are used for various purposes, such as to increase or decrease the voltage of AC power circuits, to isolate different parts of a circuit, to match the impedance of a source and a load, to transmit electrical energy over long distances, and to control the frequency and phase of AC signals .
- Transformers are passive components, meaning they do not require any external power source to operate. They are also linear components, meaning they obey the principle of superposition. However, they only work with AC signals, not with direct current (DC) signals .

Magnetic circuits

- A magnetic circuit is a closed path to which a magnetic field, represented as lines of magnetic flux, is confined.
- A magnetic circuit is analogous to an electric circuit, but instead of electric charge flowing, the magnetic flux is the quantity of interest.
- A magnetic circuit consists of a structure composed for the most part of high permeability magnetic material, such as iron, which guides the magnetic flux.
- The presence of high permeability material causes the magnetic flux to be confined to the paths defined by the structure, much as currents are confined to the conductors of an electric circuit.
- Some examples of magnetic circuits are: horseshoe magnet with iron keeper (low-reluctance circuit), horseshoe magnet with no keeper (high-reluctance circuit), electric motor (variable-reluctance circuit), and transformer (fixed-reluctance circuit).
- The main parameters of a magnetic circuit are: magnetic flux, magnetic

flux density, magnetic field intensity, magnetic potential difference, magnetic reluctance, and magnetic permeability.

- The magnetic flux is the total amount of magnetic field passing through a given area, measured in webers (Wb) or maxwells (Mx).
- The magnetic flux density is the magnetic flux per unit area, measured in teslas (T) or gauss (G).
- The magnetic field intensity is the magnetic force per unit length exerted on a unit magnetic pole, measured in amperes per meter (A/m) or oersteds (Oe).
- The magnetic potential difference is the work done in moving a unit magnetic pole around a magnetic circuit, measured in ampere-turns (At) or gilberts (Gb).
- The magnetic reluctance is the opposition to the magnetic flux in a magnetic circuit, measured in ampere-turns per weber (At/Wb) or oersteds per maxwell (Oe/Mx).
- The magnetic permeability is the measure of how easily a material can support a magnetic field, measured in henries per meter (H/m) or newtons per ampere squared (N/A^2).
- The basic laws of a magnetic circuit are: Ohm's law for magnetic circuits, Kirchhoff's laws for magnetic circuits, and Faraday's law of electromagnetic induction.
- Ohm's law for magnetic circuits states that the magnetic potential difference around a closed magnetic circuit is equal to the product of the magnetic flux and the total magnetic reluctance of the circuit.
- Kirchhoff's laws for magnetic circuits state that the sum of the magnetic potential differences around any closed loop of a magnetic circuit is zero, and the sum of the magnetic fluxes entering any node of a magnetic circuit is equal to the sum of the magnetic fluxes leaving that node.
- Faraday's law of electromagnetic induction states that the induced electromotive force (emf) in a coil is equal to the rate of change of magnetic flux through the coil.

Ideal and Practical Transformer

- A transformer is a device that transfers electrical energy from one circuit to another through mutual induction.
- An ideal transformer is a theoretical model that assumes no losses, no leakage flux, and perfect coupling between the primary and secondary windings.
- A practical transformer is a real device that has some losses, leakage flux, and imperfect coupling between the primary and secondary windings.
- The losses in a practical transformer can be classified into two types: core losses and copper losses.
- Core losses are caused by the hysteresis and eddy currents in the magnetic core of the transformer. They depend on the frequency and flux density of the alternating current.

- Copper losses are caused by the resistance of the windings. They depend on the current and the resistance of the windings.
- The leakage flux is the portion of the magnetic flux that does not link the primary and secondary windings. It reduces the mutual inductance and the efficiency of the transformer.
- The imperfect coupling between the primary and secondary windings means that the voltage ratio is not equal to the turns ratio. It also introduces a phase shift between the input and output voltages.
- The performance of a practical transformer can be analyzed using the equivalent circuit, which consists of an ideal transformer and some additional elements that represent the losses, leakage flux, and imperfect coupling.

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is the content for the topic of equivalent circuit for the notes of the Unit 3 - Transformers in the subject of FUNDAMENTALS OF ELECTRICAL ENGINEERING.

Equivalent Circuit

- An equivalent circuit is a simplified representation of a complex circuit or device that preserves its essential characteristics, such as voltage, current, power, and impedance.
- An equivalent circuit can be used to analyze the behavior and performance of a circuit or device under different conditions, such as load, frequency, and temperature.
- An equivalent circuit can also be used to compare and contrast different circuits or devices, such as transformers, motors, and generators.
- A transformer is a device that transfers electrical energy from one circuit to another by electromagnetic induction, without changing the frequency.
- A transformer consists of two or more coils of wire, called the primary and secondary windings, that are wound around a common core, usually made of iron or steel.
- The primary winding is connected to the input voltage source, and the secondary winding is connected to the output load.
- When an alternating current (AC) flows through the primary winding, it creates a varying magnetic flux in the core, which induces an electromotive force (EMF) in the secondary winding, according to Faraday's law of induction.
- The ratio of the number of turns in the primary and secondary windings, called the turns ratio, determines the ratio of the input and output voltages, according to the transformer equation:

- $V_s / V_p = N_s / N_p$
- where V_s is the secondary voltage, V_p is the primary voltage, N_s is the number of turns in the secondary winding, and N_p is the number of turns in the primary winding.
- A transformer can be classified as a step-up or a step-down transformer, depending on whether the secondary voltage is higher or lower than the primary voltage, respectively.
- A transformer can also be classified as an ideal or a practical transformer, depending on whether it has any losses or not.
- An ideal transformer is a transformer that has no losses, such as resistance, leakage flux, hysteresis, and eddy currents. It also has no magnetizing current, which is the current required to establish the magnetic flux in the core.
- An ideal transformer has the following characteristics:
 - The input power is equal to the output power, according to the principle of conservation of energy:
 - * $P_s = P_p$
 - * where P_s is the secondary power, and P_p is the primary power.
 - The input and output currents are inversely proportional to the turns ratio, according to the transformer equation:
 - * $I_s / I_p = N_p / N_s$
 - * where I_s is the secondary current, and I_p is the primary current.
 - The input and output impedances are proportional to the square of the turns ratio, according to the impedance transformation equation:
 - * $Z_s / Z_p = (N_s / N_p)^2$
 - * where Z_s is the secondary impedance, and Z_p is the primary impedance.
- An ideal transformer can be represented by an equivalent circuit that consists of an ideal voltage source in series with the primary winding, and a load resistor in parallel with the secondary winding, as shown below:
 - ideal transformer equivalent circuit
 - where V_p is the primary voltage, V_s is the secondary voltage, I_p is the primary current, I_s is the secondary current, R_L is the load resistance, and N_p and N_s are the number of turns in the primary and secondary windings, respectively.
- A practical transformer is a transformer that has some losses, such as resistance, leakage flux, hysteresis, and eddy currents. It also has some

magnetizing current, which is the current required to establish the magnetic flux in the core

Losses in Transformers

Transformers are devices that transfer electrical energy from one circuit to another by electromagnetic induction. They are used to increase or decrease the voltage and current levels of AC power systems. However, transformers are not ideal and they incur some losses during the energy conversion process. These losses reduce the efficiency and performance of the transformer. The main types of losses in transformers are:

- **Copper loss:** This is the power loss due to the resistance of the transformer windings. It is proportional to the square of the current flowing through the windings and the resistance of the windings. Copper loss can be reduced by using thicker wires or better conductors for the windings. Copper loss is also called ohmic loss, variable loss or resistive loss .
- **Core loss:** This is the power loss due to the magnetization and demagnetization of the transformer core. It consists of two components: hysteresis loss and eddy current loss. Hysteresis loss is caused by the repeated reversal of the magnetic domains in the core material as the alternating magnetic flux changes direction. Eddy current loss is caused by the induced currents in the core that circulate in small loops and oppose the main flux. Core loss can be reduced by using better core materials with low hysteresis and high resistivity, or by laminating the core into thin sheets to increase the resistance of the eddy current paths. Core loss is also called iron loss or constant loss .
- **Stray loss:** This is the power loss due to the leakage of the magnetic flux from the transformer. Some of the flux generated by the primary winding does not link with the secondary winding and escapes to the surrounding air or nearby metal objects. This flux causes unwanted heating and eddy currents in the transformer and its surroundings. Stray loss can be reduced by improving the design and construction of the transformer to minimize the air gaps and the leakage paths. Stray loss is also called leakage loss .
- **Dielectric loss:** This is the power loss due to the heating of the insulation materials used in the transformer. The insulation materials have some electrical conductivity and they dissipate some energy when subjected to the electric field of the transformer. Dielectric loss can be reduced by using better insulation materials with low conductivity and high breakdown strength. Dielectric loss is also called insulation loss.

The total loss in a transformer is the sum of the copper loss, the core loss, the stray loss and the dielectric loss. The efficiency of a transformer is the ratio of the output power to the input power, or the ratio of the input power minus the total loss to the input power. The efficiency of a transformer can be improved

by reducing the losses as much as possible.

Regulation and Efficiency of Transformers

- Regulation of a transformer is the measure of how well the output voltage of the transformer remains constant under varying load conditions.
- Regulation is expressed as a percentage of the no-load voltage, and is calculated as:

$$\% \text{ Regulation} = [(V_{NL} - V_{FL}) / V_{FL}] \times 100$$

where V_{NL} is the no-load voltage, and V_{FL} is the full-load voltage.

- A transformer with low regulation has a small change in output voltage when the load changes, and is desirable for most applications.
- A transformer with high regulation has a large change in output voltage when the load changes, and may cause problems for sensitive equipment.
- Efficiency of a transformer is the ratio of the output power to the input power, expressed as a percentage.
- Efficiency is calculated as:

$$\% \text{ Efficiency} = [(P_{out} / P_{in}) \times 100]$$

where P_{out} is the output power, and P_{in} is the input power.

- A transformer with high efficiency has a small loss of power in the form of heat, noise, and leakage flux, and is desirable for energy saving and environmental reasons.
- A transformer with low efficiency has a large loss of power in the form of heat, noise, and leakage flux, and may cause overheating, fire hazards, and increased carbon emissions.
- The efficiency of a transformer depends on several factors, such as the design, construction, material, and operating conditions of the transformer.
- The Department of Energy (DOE) has established minimum efficiency standards for different types of transformers sold in the United States, to improve the reliability of the power grid, lower the utility bills, and reduce the greenhouse gas emissions .
- The most recent of these standards are the DOE 2016 efficiency requirements, which increased the expected efficiency ratings from 98.36 to 99.49% (DOE 2010) to 98.70 to 99.55% (DOE 2016) for various categories of transformers.

Unit 4 - Electrical machines

- Electrical machines are devices that use electromagnetic forces to convert electrical energy to mechanical energy or vice versa.

- Electrical machines can be classified into three main types: transformers, generators, and motors.
- Transformers are electrical machines that transfer electrical power from one circuit to another without changing the frequency, but with a different voltage and current.
- Generators are electrical machines that convert mechanical power to electrical power by inducing an electromotive force in a coil of wire that rotates in a magnetic field.
- Motors are electrical machines that convert electrical power to mechanical power by creating a torque on a shaft that rotates in a magnetic field.
- Electrical machines can also be classified based on the type of current they use: direct current (DC) or alternating current (AC).
- DC machines are electrical machines that operate on DC power sources, such as batteries or rectifiers. DC machines can be further divided into DC motors and DC generators.
- AC machines are electrical machines that operate on AC power sources, such as the mains supply or inverters. AC machines can be further divided into synchronous machines and induction machines.
- Synchronous machines are AC machines that rotate at a constant speed that is proportional to the frequency of the AC supply. Synchronous machines can be either synchronous motors or synchronous generators (also called alternators).
- Induction machines are AC machines that rotate at a speed that is slightly less than the synchronous speed. Induction machines can be either induction motors or induction generators.
- The principles and analysis of electrical machines involve the concepts of magnetic circuits, electromechanical energy conversion, equivalent circuits, power flow, torque, speed, efficiency, and losses.
- The design of electrical machines involves the selection of materials, dimensions, winding configurations, cooling methods, and control strategies to achieve the desired performance, reliability, and cost.

DC Machines

- A DC machine is an electromechanical device that is used to convert electrical energy into mechanical energy or vice versa.
- The working principle of a DC machine is based on an effect when a current carrying conductor is placed in a magnetic field, and then the magnetic force generates a torque that rotates the DC machine.
- The main components of a DC machine are:
 - Yoke: The outer frame of the machine that provides mechanical support and magnetic flux path.
 - Poles and pole shoes: The projections on the inner side of the yoke that carry field windings and create magnetic poles.
 - Armature core: The cylindrical rotor that has slots for armature windings and commutator segments.

- Armature winding: The winding that carries the armature current and interacts with the magnetic field.
- Commutator: The device that connects the armature winding to the external circuit and reverses the direction of current periodically.
- Brushes: The conductors that slide on the commutator and transfer the current between the armature and the external circuit.
- The types of DC machines are classified based on the connection of field winding to the armature winding:
 - Separately excited DC machine: The field winding is supplied by a separate DC source.
 - Shunt-wound DC machine: The field winding is connected in parallel with the armature winding.
 - Series-wound DC machine: The field winding is connected in series with the armature winding.
 - Compound-wound DC machine: The field winding is composed of two parts, one in parallel and one in series with the armature winding.
- The applications of DC machines are:
 - DC generators: They are used to convert mechanical energy into electrical energy. They are mainly used for battery charging, electroplating, arc welding, lighting, etc.
 - DC motors: They are used to convert electrical energy into mechanical energy. They are mainly used for traction, cranes, hoists, elevators, electric vehicles, etc.

Principle and Construction of Electrical Machines

- Electrical machines are devices that convert mechanical energy to electrical energy and vice versa.
- The principle of operation of electrical machines is based on the interaction of magnetic fields and electric currents.
- The basic elements of an electrical machine are:
 - A magnetic field, which can be produced by permanent magnets or electromagnets.
 - A conductor, which can be a wire or a coil, that carries an electric current.
 - A force or torque, which is generated by the interaction of the magnetic field and the electric current.
- The direction of the force or torque can be determined by Fleming's left-hand rule for motors and Fleming's right-hand rule for generators.
- The magnitude of the force or torque depends on the strength of the magnetic field, the amount of current, and the length of the conductor.
- There are two main types of electrical machines: direct current (DC) machines and alternating current (AC) machines.
- DC machines operate on a constant voltage and current, and can be classified into DC generators and DC motors.
- DC generators convert mechanical energy to electrical energy by rotating

a coil in a magnetic field, which induces an electromotive force (EMF) in the coil.

- DC motors convert electrical energy to mechanical energy by applying a voltage and current to a coil in a magnetic field, which produces a torque on the coil.
- The construction of a DC machine consists of the following parts :
 - Yoke: The outer frame of the machine, which provides mechanical support and a low reluctance path for the magnetic flux.
 - Poles: The parts of the machine that produce the magnetic field, which can be salient or non-salient. Each pole has a pole core and a pole shoe, which are made of laminated iron, and a pole coil or field coil, which is made of copper wire and carries the field current.
 - Armature: The rotating part of the machine, which contains the armature core and the armature winding. The armature core is made of laminated iron and provides a low reluctance path for the magnetic flux. The armature winding is made of copper wire and carries the armature current. The armature winding can be lap winding or wave winding, depending on the connection of the coils.
 - Commutator: A cylindrical device that consists of copper segments insulated from each other, which are connected to the armature winding. The commutator reverses the direction of the current in the armature winding every half cycle, to maintain a unidirectional torque on the armature.
 - Brushes: Carbon or graphite blocks that slide on the commutator and provide electrical contact between the armature winding and the external circuit.
 - Bearings: Mechanical devices that support the armature shaft and reduce friction and wear.
- AC machines operate on a varying voltage and current, and can be classified into synchronous machines and induction machines.
- Synchronous machines can operate as generators or motors, depending on the direction of power flow. They have a constant speed that is proportional to the frequency of the supply voltage.
- Synchronous generators convert mechanical energy to electrical energy by rotating a field winding in a magnetic field, which induces an EMF in the armature winding.
- Synchronous motors convert electrical energy to mechanical energy by applying a voltage and current to the armature winding in a magnetic field, which produces a torque on the field winding.
- The construction of a synchronous machine consists of the following parts:
 - Stator: The stationary part of the machine, which contains the stator core and the stator winding. The stator core is made of laminated iron and provides a low reluctance path for the magnetic flux. The stator winding is made of copper wire and carries the armature current. The stator winding can be single-phase or three-phase, depending on the number of phases of the supply voltage.

- Rotor: The rotating part of the machine, which contains the rotor core and the rotor winding. The rotor core is made of laminated iron and provides a low reluctance path for the magnetic flux. The rotor winding is made of copper wire and carries the field current. The rotor winding can be salient pole or cylindrical, depending on the shape of the poles. The field current can be supplied by a DC source or by a self-excited system using slip rings and brushes.
- Slip rings and brushes: Electrical devices that provide electrical contact between the rotor winding and the external circuit. They are used only in self-excited synchronous machines, where the field current is generated by the machine

Types of Electrical Machines

Electrical machines are devices that convert electrical energy to mechanical energy or vice versa. They can be classified into two main categories: static and dynamic.

- Static electrical machines are stationary devices that do not have any moving parts. They transfer electrical energy from one circuit to another by electromagnetic induction. The most common example of a static electrical machine is a transformer, which can step up or step down the voltage and current of an alternating current (AC) source. Transformers can be single-phase or three-phase, depending on the number of windings and the connections between them. Transformers are used for power transmission, distribution, and conversion applications.
- Dynamic electrical machines are rotating devices that have a rotor and a stator. The rotor is the part that spins, while the stator is the part that remains fixed. The rotor and the stator have coils of wire that carry electric currents and create magnetic fields. The interaction between the magnetic fields of the rotor and the stator produces torque, which causes the rotor to rotate. Dynamic electrical machines can be divided into three types: direct current (DC) machines, synchronous machines, and induction machines.
 - DC machines are dynamic electrical machines that operate on direct current (DC) sources. They have permanent magnets or electromagnets in the rotor or the stator, which create a constant magnetic field. The rotor coil is supplied with current through brushes and commutators, which switch the direction of the current as the rotor rotates. This ensures that the torque is always in the same direction. DC machines can be used as generators or motors, depending on the direction of the energy conversion. DC machines are used for applications that require variable speed and high torque, such as electric vehicles, elevators, and cranes.
 - Synchronous machines are dynamic electrical machines that operate

on alternating current (AC) sources. They have electromagnets in both the rotor and the stator, which create alternating magnetic fields. The rotor is supplied with current through slip rings and brushes, which do not switch the direction of the current. The rotor and the stator rotate at the same speed, which is called the synchronous speed. The synchronous speed depends on the frequency of the AC source and the number of poles in the machine. Synchronous machines can be used as generators or motors, depending on the direction of the energy conversion. Synchronous machines are used for applications that require constant speed and high efficiency, such as power plants, pumps, and compressors.

- Induction machines are dynamic electrical machines that operate on alternating current (AC) sources. They have electromagnets in the stator, but not in the rotor. The rotor is made of conductive bars or coils, which are short-circuited at the ends. The alternating magnetic field of the stator induces currents in the rotor, which create a magnetic field that opposes the stator field. This causes the rotor to rotate, but at a lower speed than the stator. The difference between the rotor speed and the stator speed is called the slip. Induction machines can be used as motors only, not as generators. Induction machines are the most common type of electrical machines, as they are simple, robust, and cheap. They are used for applications that require variable speed and moderate torque, such as fans, blowers, and pumps.

EMF equation of generator and torque equation of motor

- A generator is a device that converts mechanical energy into electrical energy by inducing a voltage in a coil of wire that rotates in a magnetic field.
- A motor is a device that converts electrical energy into mechanical energy by applying a torque to a coil of wire that rotates in a magnetic field.
- The EMF equation of a generator relates the generated voltage (E_g) to the number of turns (N), the magnetic flux (Φ), the speed of rotation (Z), the number of poles (P), and the number of parallel paths (A) in the coil. The equation is:
 - $E_g = \Phi Z N / 60 A$
- The EMF equation of a motor relates the back EMF (E_b) to the same parameters as the generator. The back EMF is the voltage induced in the coil that opposes the applied voltage and limits the current in the motor. The equation is:
 - $E_b = \Phi Z N / 60 A$

- The torque equation of a motor relates the torque (T) to the current (I), the magnetic flux (Φ), the number of turns (N), and the number of parallel paths (A) in the coil. The equation is:

$$- T = \Phi NI / A$$

- The torque equation of a generator is the same as the motor, except that the current is the output current of the generator, not the input current. The equation is:

$$- T = \Phi NI / A$$

- The power equation of a generator relates the output power (P) to the generated voltage (E_g) and the output current (I). The equation is:

$$- P = E_g I$$

- The power equation of a motor relates the input power (P) to the applied voltage (V) and the input current (I). The equation is:

$$- P = VI$$

- The efficiency of a generator or a motor is the ratio of the output power to the input power. The equation is:

$$- \eta = P_{out} / P_{in}$$

- The efficiency of a generator or a motor depends on the losses due to friction, heat, and resistance in the coil. The higher the losses, the lower the efficiency.

Applications of DC motors (simple numerical problems)

DC motors are electric motors that are powered by direct current (DC), such as from a battery or DC power supply. They have different types, features and applications depending on their design and construction. Here are some of the common applications of DC motors:

- **Computer equipment:** DC motors are used for CPU cooling fans and the drive motors for HDDs and CD-ROM drives. They provide low noise, high speed and precise control.
- **Audio and video equipment:** DC motors are also used in audio CD, DVD, and Blu-ray players. They offer smooth rotation, low vibration and accurate positioning.
- **Home appliances:** DC motors are used in various home appliances, such as vacuum cleaners, sewing machines, electric fans, hair dryers, etc. They provide high torque, variable speed and low power consumption.
- **Automobiles:** DC motors are used in many parts of automobiles, such as wiper motors, power seat motors, power window motors, etc. They

offer high reliability, durability and efficiency.

- **Industrial machinery and medical equipment:** DC motors are used as servo motors in industrial robots, CNC machines, etc. They provide high precision, fast response and wide range of speed. DC motors are also used in fan motors in respirators and oxygen concentrators. They offer low noise, long life and stable operation.

Depending on the type of DC motor, the applications may vary. There are four main types of DC motors: permanent magnet, series, shunt and compound.

- **Permanent magnet DC motor:** This type of motor has a permanent magnet to create the magnetic field. It has a simple structure, low cost and high efficiency. It is suitable for low power and constant speed applications, such as toys, electric toothbrushes, etc.
- **Series DC motor:** This type of motor has the field winding connected in series with the armature winding. It has a high starting torque and variable speed. It is suitable for high power and variable load applications, such as traction systems, cranes, air compressors, etc.
- **Shunt DC motor:** This type of motor has the field winding connected in parallel with the armature winding. It has a low starting torque and constant speed. It is suitable for low power and constant load applications, such as fans, blowers, pumps, etc.
- **Compound DC motor:** This type of motor has both series and shunt field windings. It has a combination of characteristics of both series and shunt motors. It can be adjusted for different speed and torque requirements. It is suitable for medium power and variable load applications, such as elevators, lathes, printing presses, etc.

To solve simple numerical problems involving DC motors, we need to know some basic formulas and concepts, such as:

- **Back emf:** This is the voltage induced in the armature winding due to the rotation of the armature in the magnetic field. It opposes the applied voltage and limits the current in the circuit. It is given by:

$$E_b = \frac{P\Phi ZN}{60A}$$

where E_b is the back emf, P is the number of poles, Φ is the flux per pole, Z is the number of armature conductors, N is the speed in rpm, and A is the number of parallel paths in the armature.

- **Torque:** This is the rotational force produced by the motor. It is proportional to the product of the armature current and the flux per pole. It is given by:

$$T = \frac{P\Phi I_a}{2A}$$

where T is the torque, I_a is the armature current.

- **Power:** This is the rate of doing work by the motor. It is proportional to the product of the torque and the speed. It is given by:

$$P = 2 \pi NT / 60$$

where P is the power.

- **Efficiency:** This is the ratio of the output power to the input power of

Three Phase Induction Motor

- A three phase induction motor is a type of AC motor that uses three alternating currents (AC) to generate a rotating magnetic field in the stator .
- The rotating magnetic field induces an electromotive force (EMF) in the rotor, which causes the rotor to rotate and produce mechanical power .
- The rotor speed is always less than the synchronous speed of the stator magnetic field. The difference in speed is called the slip .
- The three phase induction motor is the most widely used electric motor in industry, because of its simplicity, robustness, high efficiency, and low cost .
- There are two main types of three phase induction motors: squirrel cage induction motor and slip-ring or wound rotor induction motor .
- The squirrel cage induction motor has a rotor that consists of a cylindrical laminated core with bars of conductive material embedded in slots. The bars are short-circuited at both ends by end rings .
- The slip-ring or wound rotor induction motor has a rotor that consists of a cylindrical laminated core with coils of insulated wire wound around it. The coils are connected to slip rings mounted on the rotor shaft, which allow external resistors or other devices to be connected to the rotor circuit .
- The advantages of the squirrel cage induction motor are: simple and rugged construction, low maintenance, high efficiency, low cost, and self-starting capability .
- The advantages of the slip-ring or wound rotor induction motor are: better starting performance, higher starting torque, lower starting current, and speed control capability .
- The disadvantages of the squirrel cage induction motor are: low starting torque, high starting current, and poor speed control capability .
- The disadvantages of the slip-ring or wound rotor induction motor are: complex and expensive construction, high maintenance, lower efficiency, and higher losses .
- The three phase induction motor can be controlled by varying the frequency, voltage, or slip of the stator or rotor supply.
- The three phase induction motor can be used for various applications, such as pumps, fans, compressors, conveyors, lifts, cranes, mills, mixers, and

generators .

Principle and Construction of Electrical Machines

- Electrical machines are devices that convert mechanical energy to electrical energy and vice versa.
- The principle of operation of electrical machines is based on the interaction of magnetic fields and electric currents.
- The magnetic field can be produced by permanent magnets or electromagnets (coils carrying current).
- The electric current can be induced by relative motion between a conductor and a magnetic field, or supplied by an external source.
- The mechanical force or torque on the conductor is due to the Lorentz force law, which states that a current-carrying conductor in a magnetic field experiences a force perpendicular to both the current and the field.
- The direction of the force or torque can be determined by the right-hand rule or Fleming's left-hand rule.
- The magnitude of the force or torque depends on the strength of the current, the magnetic field, and the angle between them.
- There are two main types of electrical machines: generators and motors.
- Generators convert mechanical energy to electrical energy by inducing an electromotive force (emf) or voltage in a conductor that moves in a magnetic field.
- Motors convert electrical energy to mechanical energy by applying a current to a conductor that is placed in a magnetic field, causing it to rotate.
- Generators and motors can be classified into direct current (DC) and alternating current (AC) machines, depending on the type of current they produce or use.
- DC machines have a constant polarity of voltage and current, while AC machines have a sinusoidal or alternating polarity of voltage and current.
- The construction of electrical machines consists of some common components, such as:
 - Yoke: The outer frame or casing that supports and protects the machine. It is usually made of cast iron or steel.
 - Poles: The parts that produce or receive the magnetic field. They are attached to the yoke and have pole cores and pole shoes. The pole cores are made of laminated steel and carry the field coils or windings. The pole shoes are extensions of the pole cores that increase the area of contact with the armature and reduce the reluctance of the magnetic circuit.
 - Armature: The part that carries the armature windings or conductors. It is usually a cylindrical drum or disc that rotates in the magnetic field. The armature core is made of laminated steel and has slots for the armature windings. The armature windings are connected in series or parallel to form loops or coils. The armature conductors are connected to the commutator or slip rings, depending

on the type of machine.

- Commutator: A cylindrical device that consists of insulated copper segments or bars that are attached to the armature conductors. It rotates with the armature and acts as a mechanical rectifier or switch that reverses the direction of the current in the armature windings every half cycle, to maintain a unidirectional torque in DC machines.
- Slip rings: Two or more metal rings that are fixed to the armature shaft and insulated from each other. They rotate with the armature and provide a continuous connection between the armature windings and the external circuit in AC machines.
- Brushes: Two or more carbon or metal blocks that are pressed against the commutator or slip rings by springs. They slide over the commutator or slip rings and collect or supply the current to or from the armature windings. They are connected to the terminals of the machine.
- Bearings: The parts that support the armature shaft and allow it to rotate smoothly and with minimum friction. They are usually ball or roller bearings that are lubricated and sealed.

Types of Electrical Machines

Electrical machines are devices that convert electrical energy into mechanical energy or vice versa. They can be classified into two main categories: static and dynamic.

- Static electrical machines are stationary devices that do not have any moving parts. They transfer electrical energy from one circuit to another without changing its frequency or waveform. The most common example of a static electrical machine is a transformer, which can step up or step down the voltage and current of an alternating current (AC) source. Transformers are used for power transmission, distribution, and isolation purposes.
- Dynamic electrical machines are rotating devices that have a rotor and a stator. They can either generate electrical energy from mechanical energy (generators) or consume electrical energy to produce mechanical energy (motors). The most common types of dynamic electrical machines are:
 - DC machines: These machines use direct current (DC) as the input or output. They have a commutator and brushes that switch the polarity of the current in the rotor coil to maintain a constant torque. DC machines can be classified into permanent magnet DC machines, which have permanent magnets in the rotor, and wound-field DC machines, which have electromagnets in the rotor. DC machines are used for applications that require variable speed, high torque, and precise control.
 - Synchronous machines: These machines use alternating current (AC)

as the input or output. They have a constant speed that is proportional to the frequency of the AC source. They have a rotor that is magnetized by either permanent magnets or electromagnets, and a stator that has three-phase windings. Synchronous machines can be classified into synchronous generators, which produce AC power from mechanical energy, and synchronous motors, which consume AC power to produce mechanical energy. Synchronous machines are used for applications that require high power, high efficiency, and constant speed.

- Induction machines: These machines use alternating current (AC) as the input or output. They have a variable speed that depends on the load and the slip between the rotor and the stator. They have a rotor that is either a squirrel-cage or a wound-rotor, and a stator that has three-phase windings. Induction machines can be classified into induction generators, which produce AC power from mechanical energy, and induction motors, which consume AC power to produce mechanical energy. Induction machines are used for applications that require low cost, low maintenance, and high reliability.

Slip-torque characteristics of induction motor

- The slip-torque characteristic of an induction motor is the relationship between the torque produced by the motor and the slip of the rotor with respect to the synchronous speed.
- The slip of the rotor is defined as the difference between the synchronous speed and the actual speed of the rotor, expressed as a fraction of the synchronous speed.
- The slip-torque characteristic can be derived from the equivalent circuit of the induction motor, by equating the mechanical power output to the electrical power input to the rotor.
- The torque-slip characteristic curve can be divided roughly into three regions :
 - Low slip region: This is the region where the slip is very small and the rotor speed is near the synchronous speed. In this region, the torque is proportional to the slip and the rotor resistance. The torque is low and increases linearly with the slip. The power factor and the efficiency are high in this region.
 - Medium slip region: This is the region where the slip is moderate and the rotor speed is lower than the synchronous speed. In this region, the torque is proportional to the square of the slip and inversely proportional to the rotor reactance. The torque is higher and reaches a maximum value at a certain slip, called the pull-out slip or the critical slip. The power factor and the efficiency are lower in this region.
 - High slip region: This is the region where the slip is high and the rotor

speed is much lower than the synchronous speed. In this region, the torque is inversely proportional to the slip and the rotor resistance. The torque is low and decreases rapidly with the slip. The power factor and the efficiency are very low in this region.

- The shape and the position of the torque-slip characteristic curve depend on the parameters of the induction motor, such as the rotor resistance, the rotor reactance, the stator resistance, the stator reactance, the magnetizing reactance, and the supply voltage and frequency.
- The torque-slip characteristic curve can be modified by changing the rotor resistance, either by inserting external resistors in the rotor circuit or by using a wound rotor with slip rings and brushes. By increasing the rotor resistance, the maximum torque is increased and the pull-out slip is shifted to a higher value. This improves the starting torque and the speed regulation of the motor, but reduces the efficiency and the power factor.

Applications (Numerical problems related to slip only)

- Slip is the difference between the synchronous speed and the actual speed of an induction motor, expressed as a fraction or percentage of the synchronous speed.
- Slip can be calculated by the formula: $s = (N_s - N)/N_s$, where s is the slip, N_s is the synchronous speed, and N is the actual speed of the motor.
- Slip can also be expressed in terms of frequency as: $s = (f_s - f_r)/f_s$, where f_s is the supply frequency, and f_r is the rotor frequency.
- Slip is an important parameter for induction motors, as it affects the torque, power, efficiency, and speed regulation of the motor.
- Some numerical problems related to slip are:
 - Example 1: A three-phase induction motor has a synchronous speed of 1200 rpm and runs at 1140 rpm. Find the slip of the motor.
 - * Solution: Using the formula $s = (N_s - N)/N_s$, we get:
 - $s = (1200 - 1140)/1200$
 - $s = 0.05$
 - The slip of the motor is 0.05 or 5%.
 - Example 2: A four-pole induction motor is connected to a 50 Hz supply and has a full-load slip of 4%. Find the full-load speed of the motor.
 - * Solution: The synchronous speed of a four-pole motor is given by: $N_s = 120f/p$, where f is the supply frequency, and p is the number of poles. Substituting the values, we get:
 - $N_s = 120 \times 50 / 4$
 - $N_s = 1500 \text{ rpm}$
 - The full-load speed of the motor is given by: $N = N_s (1 - s)$, where s is the slip. Substituting the values, we get:

- $N = 1500 (1 - 0.04)$
- $N = 1440 \text{ rpm}$
- The full-load speed of the motor is 1440 rpm.
- Example 3: A six-pole induction motor has a rotor frequency of 2 Hz when the supply frequency is 60 Hz. Find the slip and the speed of the motor.
 - * Solution: The slip of the motor is given by: $s = (f_s - f_r)/f_s$, where f_s is the supply frequency, and f_r is the rotor frequency. Substituting the values, we get:
 - $s = (60 - 2)/60$
 - $s = 0.9667$
 - The slip of the motor is 0.9667 or 96.67%.
 - The synchronous speed of a six-pole motor is given by: $N_s = 120f/p$, where f is the supply frequency, and p is the number of poles. Substituting the values, we get:
 - $N_s = 120 \times 60 / 6$
 - $N_s = 1200 \text{ rpm}$
 - The speed of the motor is given by: $N = N_s (1 - s)$, where s is the slip. Substituting the values, we get:
 - $N = 1200 (1 - 0.9667)$
 - $N = 40 \text{ rpm}$
 - The speed of the motor is 40 rpm.

Single Phase Induction Motor

- A single phase induction motor is a type of AC motor that operates on single phase power supply .
- It has a stator with distributed single phase winding and a rotor with a cage winding .
- The stator and the rotor have a uniform air gap, but no electrical connection.
- The stator produces a pulsating magnetic field, rather than a rotating one as in a three phase motor .
- This means that the single phase induction motor has no starting torque and cannot start by itself.
- To make the motor self-starting, various methods are used to create a rotating magnetic field at least at the start, such as:
 - Split-phase method: The stator has two windings, one main and one auxiliary, with different resistances and inductances. The auxiliary winding is connected to a centrifugal switch or a relay that disconnects it when the motor reaches a certain speed. The phase difference between the two windings creates a rotating field.
 - Capacitor-start method: The stator has two windings, one main and one auxiliary, with a capacitor connected in series with the auxiliary winding. The capacitor creates a phase shift between the two windings, resulting in a rotating field. The auxiliary winding is dis-

connected by a switch or a relay when the motor reaches a certain speed.

- Permanent-split capacitor method: The stator has two windings, one main and one auxiliary, with a capacitor connected in series with the auxiliary winding. The capacitor remains in the circuit permanently, creating a phase shift between the two windings and a rotating field. This method has a higher efficiency and a lower noise than the capacitor-start method .
- Shaded-pole method: The stator has a single winding with a part of each pole shaded by a copper ring or a short-circuited coil. The shaded part of the pole creates a phase shift and a rotating field. This method is simple and cheap, but has a low power factor and a low efficiency.
- Once the motor is started, the induced currents in the rotor create a magnetic field that interacts with the stator field and produces a torque, and thus rotation.
- The speed of the single phase induction motor depends on the frequency of the supply voltage and the number of poles in the stator. The slip, which is the difference between the synchronous speed and the actual speed, is usually small.
- The single phase induction motor is widely used in domestic and industrial applications, such as fans, blowers, pumps, compressors, refrigerators, washing machines, etc.

Principle of operation and introduction to methods of starting for the notes of the Unit 4 - Electrical machines in the subject of FUNDAMENTALS OF ELECTRICAL ENGINEERING

- Electrical machines are devices that convert electrical energy into mechanical energy or vice versa. They can be classified into two main categories: direct current (DC) machines and alternating current (AC) machines.
- DC machines operate on the principle of electromagnetic induction, which states that when a current-carrying conductor is placed in a magnetic field, it experiences a force that tends to rotate it. This force is called torque and it is proportional to the current, the magnetic field and the angle between them. DC machines can be further divided into DC generators and DC motors, depending on whether they produce or consume electrical energy.
- DC generators convert mechanical energy into electrical energy by rotating a coil of wire in a magnetic field. The induced voltage in the coil depends on the number of turns, the magnetic flux and the speed of rotation. DC generators can be classified into three types: separately excited, self-excited and permanent magnet. Separately excited generators have an independent source of field current, while self-excited generators use part of the output voltage to supply the field current. Permanent magnet generators have permanent magnets instead of electromagnets to provide

the magnetic field.

- DC motors convert electrical energy into mechanical energy by applying a voltage to a coil of wire that rotates in a magnetic field. The direction of rotation depends on the polarity of the voltage and the magnetic field. DC motors can be classified into three types: shunt, series and compound. Shunt motors have a constant field current, while series motors have a variable field current that depends on the load. Compound motors have a combination of shunt and series field windings.
- AC machines operate on the principle of rotating magnetic field, which states that a set of coils carrying alternating currents can produce a magnetic field that rotates at a constant speed. The speed of rotation depends on the frequency of the currents and the number of poles in the coils. AC machines can be further divided into synchronous machines and induction machines, depending on whether they have a fixed or a variable speed of rotation.
- Synchronous machines have a rotor that rotates at the same speed as the rotating magnetic field. The rotor can be either a permanent magnet or an electromagnet. Synchronous machines can be used as generators or motors, depending on whether they produce or consume electrical energy. Synchronous generators are commonly used in power plants to generate electricity, while synchronous motors are used in applications that require precise speed control, such as clocks and robots.
- Induction machines have a rotor that rotates at a slightly lower speed than the rotating magnetic field. The rotor can be either a squirrel cage or a wound rotor. Induction machines can only be used as motors, as they cannot generate electricity. Induction motors are the most widely used type of electric motors, as they are simple, cheap, reliable and efficient. They are used in various applications, such as fans, pumps, compressors, conveyors and elevators.
- Starting methods for electrical machines are the techniques used to initiate the operation of the machines. They are necessary because the machines may not be able to start under normal conditions, due to high starting currents, low starting torques, mechanical stresses or other factors. Starting methods can be classified into two types: full voltage and reduced voltage methods.
- Full voltage methods apply the rated voltage to the machine terminals at the start. They are simple and cheap, but they may cause high starting currents, low power factor, voltage dips and mechanical shocks. Full voltage methods are suitable for small and lightly loaded machines, such as shunt and compound DC motors, permanent magnet DC motors, permanent magnet synchronous motors and squirrel cage induction motors.
- Reduced voltage methods apply a lower voltage than the rated voltage to the machine terminals at the start. They are more complex and expensive, but they can reduce the starting currents, improve the power factor, limit the voltage dips and minimize the mechanical stresses. Reduced voltage methods are suitable for large and heavily loaded machines, such as series

DC motors, wound rotor DC motors, wound rotor synchronous motors and wound rotor induction motors.

- Some examples of reduced voltage methods are: star-delta starter, auto-transformer starter, resistor starter, reactor starter, soft starter and variable frequency drive. Each of these methods has its own advantages and disadvantages, depending on the type and size of the machine, the load characteristics, the supply conditions and the desired performance.

Applications of Electrical Machines

Electrical machines are devices that convert electrical energy into mechanical energy or vice versa. They are widely used in various fields of engineering, industry, and everyday life. Some of the applications of electrical machines are:

- **Electric motors** are used to drive various machines, such as fans, pumps, compressors, conveyors, cranes, elevators, robots, electric vehicles, etc. Electric motors can be classified into two main types: AC motors and DC motors. AC motors operate on alternating current and can be further divided into synchronous motors and induction motors. DC motors operate on direct current and can be further divided into permanent magnet motors, shunt motors, series motors, and compound motors .
- **Electric generators** are used to produce electrical energy from mechanical energy, such as in power plants, wind turbines, hydroelectric dams, etc. Electric generators can also be classified into two main types: AC generators and DC generators. AC generators produce alternating current and are also called alternators. They can be further divided into synchronous generators and induction generators. DC generators produce direct current and are also called dynamos. They can be further divided into permanent magnet generators, shunt generators, series generators, and compound generators .
- **Electric transformers** are used to change the voltage and current levels of an alternating current, such as in power transmission and distribution systems, electric appliances, etc. Electric transformers can be classified into two main types: step-up transformers and step-down transformers. Step-up transformers increase the voltage and decrease the current, while step-down transformers decrease the voltage and increase the current.
- **Electric actuators** are used to control the motion of mechanical systems, such as valves, switches, relays, etc. Electric actuators can be classified into two main types: linear actuators and rotary actuators. Linear actuators produce linear motion, while rotary actuators produce rotational motion. Electric actuators can use different types of electric motors, such as stepper motors, servo motors, solenoid motors, etc.

Three Phase Synchronous Machines

- A three phase synchronous machine is a type of electric machine that can operate as either a generator or a motor, depending on the direction of

power flow.

- A three phase synchronous machine consists of two main parts: a stator and a rotor.
- The stator is the stationary part of the machine that contains a three phase winding, which is connected to the AC supply or the load. The stator winding produces a rotating magnetic field when energized by AC current.
- The rotor is the rotating part of the machine that contains a DC field winding, which is excited by a DC source or an exciter. The rotor field interacts with the stator field to produce torque and power.
- The rotor can be either cylindrical (round rotor) or salient pole (projected pole), depending on the shape and distribution of the field poles. Round rotor machines are used for high speed and high power applications, such as steam turbines and gas turbines. Salient pole machines are used for low speed and low power applications, such as hydro generators and single phase motors.
- The speed of the rotor is equal to the speed of the stator field, which is determined by the frequency of the AC supply and the number of poles on the stator. This speed is called the synchronous speed, and it is given by the formula:

$$N_s = 120f / P$$

where N_s is the synchronous speed in rpm, f is the frequency in Hz, and P is the number of poles on the stator.

- A three phase synchronous machine can operate in two modes: synchronous mode and asynchronous mode. In synchronous mode, the rotor speed is equal to the synchronous speed, and the machine is said to be in synchronism. In asynchronous mode, the rotor speed is different from the synchronous speed, and the machine is said to be out of synchronism.
- A three phase synchronous generator converts mechanical energy into electrical energy by rotating the rotor field in the same direction as the stator field. The output voltage and frequency of the generator depend on the speed and excitation of the rotor. The generator can supply power to a balanced or unbalanced load, depending on the connection of the stator winding (star or delta).
- A three phase synchronous motor converts electrical energy into mechanical energy by rotating the rotor field in the opposite direction as the stator field. The input voltage and frequency of the motor determine the speed and torque of the rotor. The motor can be started by various methods, such as damper winding, induction motor, or external source. The motor can operate at constant speed or variable speed, depending on the control of the rotor excitation.

Principle of operation of alternator and synchronous motor

- An **alternator** or **synchronous generator** is a device that converts mechanical energy into electrical energy by producing alternating current (AC) voltage.
- A **synchronous motor** is a device that converts electrical energy into mechanical energy by rotating a shaft at a constant speed that is synchronized with the frequency of the AC supply.
- Both devices work on the principle of **electromagnetic induction**, i.e., when the flux linking a conductor changes, an EMF is induced in the conductor .
- The main components of both devices are a **stator** and a **rotor**. The stator is the stationary part that contains the armature winding, and the rotor is the rotating part that contains the field winding.
- The stator winding is connected to a three-phase AC supply, which creates a rotating magnetic field (RMF) that rotates at a speed of $120f/p$ revolutions per minute, where f is the frequency of the supply and p is the number of poles .
- The rotor winding is connected to a direct current (DC) supply, which creates a constant magnetic field that aligns with the RMF of the stator.
- In an alternator, the rotor is driven by a prime mover, such as a turbine or an engine, which causes the rotor magnetic field to cut the stator winding and induce an AC voltage in it. The output voltage of the alternator depends on the speed of the rotor, the number of turns in the stator winding, and the strength of the rotor magnetic field.
- In a synchronous motor, the stator winding is connected to a load, such as a fan or a pump, which causes the stator magnetic field to exert a torque on the rotor and make it rotate. The speed of the rotor is equal to the speed of the stator RMF, and the power factor of the motor depends on the angle between the rotor and stator magnetic fields.

Applications of Electrical Machines

Electrical machines are devices that convert electrical energy into mechanical energy or vice versa. They are classified into two main categories: rotating and linear machines. Rotating machines include motors and generators, while linear machines include actuators and transformers. Some of the applications of electrical machines are:

- **Electric motors** are used to drive various machines and devices, such as industrial fans, blowers, pumps, machine tools, household appliances, power tools, and disk drives. They can be powered by direct current (DC) or alternating current (AC), which leads to different types of motors, such as DC motors, AC motors, synchronous motors, and induction motors.
- **Electric generators** are used to produce electrical energy from mechanical energy, such as in power plants, wind turbines, hydroelectric dams, and diesel generators. They can also be powered by DC or AC, which leads

to different types of generators, such as DC generators, AC generators, synchronous generators, and induction generators.

- **Electric actuators** are used to convert electrical energy into linear motion, such as in valves, pumps, switches, and robots. They can be based on different principles, such as electromagnetic, piezoelectric, electrostatic, or shape memory alloys.
- **Electric transformers** are used to change the voltage and current levels of an AC supply, such as in power transmission, distribution, and conversion. They can be classified into different types, such as step-up, step-down, isolation, autotransformer, and current transformer.

Some of the factors that determine the choice of electrical machines for different applications are:

- **Power rating** - the amount of power that the machine can deliver or consume.
- **Speed** - the rotational or linear velocity of the machine.
- **Torque** - the rotational or linear force that the machine can produce or withstand.
- **Efficiency** - the ratio of output power to input power of the machine.
- **Cost** - the initial and operational expenses of the machine.
- **Reliability** - the ability of the machine to perform without failures or faults.
- **Maintenance** - the frequency and complexity of servicing the machine.
- **Environment** - the physical and operational conditions that the machine is exposed to, such as temperature, humidity, dust, noise, and vibration .

Unit 5 - Electrical Installations

This unit covers the following topics:

- Electrical symbols and diagrams
- Electrical circuits and components
- Electrical safety and regulations
- Electrical measurements and calculations
- Electrical installation and testing

Electrical symbols and diagrams

- Electrical symbols are used to represent different electrical devices and components in a circuit diagram.
- Electrical diagrams are graphical representations of electrical circuits that show the connections and arrangements of the components.
- There are different types of electrical diagrams, such as schematic diagrams, wiring diagrams, and layout diagrams.
- Schematic diagrams show the logic and function of a circuit using standard symbols and conventions.

- Wiring diagrams show the physical connections and layout of the wires and devices in a circuit.
- Layout diagrams show the location and orientation of the components and devices in a circuit.

Electrical circuits and components

- An electrical circuit is a closed path that allows electric current to flow from a source of voltage to a load.
- A source of voltage is a device that provides electrical energy to a circuit, such as a battery, a generator, or a solar panel.
- A load is a device that consumes electrical energy in a circuit, such as a lamp, a motor, or a heater.
- A switch is a device that can open or close a circuit, allowing or stopping the current flow.
- A resistor is a device that limits the current flow in a circuit, producing heat and voltage drop.
- A capacitor is a device that stores electrical energy in a circuit, producing voltage and current changes.
- An inductor is a device that opposes the change of current in a circuit, producing magnetic fields and voltage spikes.
- A diode is a device that allows current to flow in one direction only, blocking the reverse current.
- A transistor is a device that can amplify or switch electrical signals in a circuit, acting as a current-controlled switch.
- A LED is a device that emits light when current flows through it, acting as a light source and an indicator.

Electrical safety and regulations

- Electrical safety is the prevention of electric shock, fire, or explosion caused by electrical hazards in a circuit or an installation.
- Electrical hazards are the potential sources of danger or harm from electricity, such as exposed wires, faulty insulation, overloaded circuits, or wet conditions.
- Electrical regulations are the rules and standards that govern the design, installation, operation, and maintenance of electrical systems and equipment, such as the National Electrical Code (NEC) in the US, or the British Standard (BS) in the UK.
- Electrical regulations ensure the safety, efficiency, and reliability of electrical systems and equipment, as well as the protection of people, property, and environment from electrical hazards.
- Electrical regulations specify the requirements for electrical wiring, grounding, protection, testing, labeling, and documentation of electrical systems and equipment.

Electrical measurements and calculations

- Electrical measurements are the methods and instruments used to measure the electrical quantities in a circuit, such as voltage, current, resistance, power, energy, and frequency.
- Electrical calculations are the mathematical operations and formulas used to determine the electrical quantities in a circuit, such as Ohm's law, Kirchhoff's laws, power formula, energy formula, and frequency formula.
- Ohm's law states that the voltage across a resistor is equal to the product of the current through it and its resistance, or $V = IR$.
- Kirchhoff's current law (KCL) states that the sum of the currents entering a node in a circuit is equal to the sum of the currents leaving that node, or $I_{in} = I_{out}$.
- Kirchhoff's voltage law (KVL) states that the sum of the voltages around a closed loop in a circuit is equal to zero, or $V_{loop} = 0$.
- Power formula states that the power consumed or delivered by a device in a circuit is equal to the product of the voltage across it and the current through it, or $P = VI$.
- Energy formula states that the energy consumed or delivered by a device in a circuit is equal to the product of the power and the time, or $E = Pt$.
- Frequency formula states that the frequency of an alternating current (AC) or a periodic signal is equal to the inverse of the time period, or $f = 1/T$.

Electrical installation and testing

- Electrical installation is the process of planning, designing, installing, and connecting the electrical systems and equipment in a building or a facility, such as lighting, power, communication, and control systems.
- Electrical testing is the process of inspecting, measuring, and verifying the performance, functionality, and safety of the electrical systems and equipment in a building or a facility, using various tools and instruments, such as multimeters, oscilloscopes, and testers.
- Electrical installation and testing follow the electrical regulations and

Introduction of Switch Fuse Unit (SFU) for the notes of the Unit 5 - Electrical Installations in the subject of FUNDAMENTALS OF ELECTRICAL ENGINEERING

- A switch fuse unit (SFU) is a type of low voltage switchgear that combines a switch and a fuse in a single unit.
- SFU is used to protect and isolate electrical circuits from short circuits, overloads, and faults.
- SFU has one switch unit and one fuse unit. When the switch is operated, the contacts will get closed or opened, and then the supply will pass through or disconnect from the fuse unit to the output.
- SFU is also known as an ironclad switch, being made of iron. It may be double pole for controlling single phase two-wire circuits or triple pole for

controlling three-phase, 3-wire circuits or triple pole with neutral link for controlling 3-phase, 4-wire circuits.

- The respective switches are known as double pole ironclad (DPIC), triple pole ironclad (TPIC), and triple pole with neutral link ironclad (TPNIC) switches.
- SFU can be mounted on a panel board, a distribution board, or a wall. It has a handle for manual operation and an indicator for showing the status of the switch and the fuse.
- SFU usually uses high rupturing capacity (HRC) fuses as back up fuses in this kind of units. The fuse rating can be selected based on the connected load and should be less than the switch rating. For example, a 32A fuse can be used for a 63A rating SFU unit.
- SFU provides a reliable and economical way of protecting and controlling electrical circuits in low voltage applications. It also reduces the maintenance and replacement costs of the switch and the fuse.

MCB for the notes of the Unit 5 - Electrical Installations in the subject of FUNDAMENTALS OF ELECTRICAL ENGINEERING

- MCB stands for **Miniature Circuit Breaker**, which is an automatically operated electrical switch that protects an electrical circuit from damage caused by excess current from an overload or short circuit.
- MCBs are designed to **trip** (interrupt the current flow) after a certain amount of time depending on the overcurrent level. This prevents the circuit from overheating and causing fire or damage to the equipment.
- MCBs are usually classified by their **current rating, tripping characteristics, and breaking capacity**. The current rating is the maximum current that the MCB can carry without tripping. The tripping characteristics are the time and current values that determine when the MCB will trip. The breaking capacity is the maximum current that the MCB can safely interrupt without damage.
- MCBs are commonly used in low-voltage electrical networks, such as residential, commercial, and industrial applications. They are often installed in **distribution boards** or **consumer units** that divide the main power supply into separate circuits for different loads.
- To install an MCB, the following steps are recommended :
 - Step 1: Identify the spot to install the MCB. Turn off the power supply to the electric panel and look for an unused location to accommodate the MCB. If the location has a knockout plate, remove it before the installation.
 - Step 2: Place the MCB in the panel. Select the correct MCB that is allowed to be installed in the panel and place it next to an existing MCB. Align the clips on the back of the MCB with the slots on the panel and snap it into place.
 - Step 3: Connect the wires to the MCB. Strip the insulation from the wires and connect them to the terminals of the MCB. The hot wire

(usually black or red) goes to the terminal marked “L” or “LINE”. The neutral wire (usually white or blue) goes to the terminal marked “N” or “LOAD”. Tighten the screws to secure the wires.

- Step 4: Test the MCB. Turn on the power supply to the electric panel and check if the MCB is working properly. Use a test light or meter to verify that the MCB is receiving and delivering power. If the MCB trips, check for any wiring errors or faulty devices on the circuit.

ELCB

- ELCB stands for Earth Leakage Circuit Breaker. It is a safety device that detects small stray voltages on the metal enclosures of electrical equipment and interrupts the circuit if a dangerous voltage is detected .
- ELCB is used in electrical installations with high earth impedance to prevent shock and fire hazards from leakage currents .
- ELCB is a type of latching relay that has the incoming mains power connected through its switching contacts. When earth leakage is detected, the ELCB disconnects the power supply to the installation it protects .
- ELCB detects fault currents from live to the earth wire within the installation. It does not detect faults from live to neutral or live to live wires .
- ELCB can be classified into two types: voltage-operated ELCB and current-operated ELCB .
 - Voltage-operated ELCB measures the voltage difference between the earth wire and a separate earth electrode. It trips when the voltage exceeds a preset value. It is sensitive to the condition of the earth electrode and the soil resistivity .
 - Current-operated ELCB measures the current difference between the live and neutral wires. It trips when the current difference exceeds a preset value. It is independent of the earth electrode and the soil resistivity. It is also known as residual current device (RCD) or ground fault circuit interrupter (GFCI) .
- ELCB has some advantages and disadvantages .
 - Advantages:
 - * It provides protection against electrical shock and fire hazards from earth leakage currents.
 - * It can detect faults that are not detected by conventional circuit breakers or fuses.
 - * It can reduce the risk of electrocution and damage to equipment and property.
 - Disadvantages:
 - * It can cause nuisance tripping due to external factors such as lightning, moisture, or capacitive coupling.
 - * It can be affected by the quality and condition of the earth electrode and the soil resistivity (for voltage-operated ELCB).

- * It can be bypassed by connecting the earth wire to the neutral wire or by removing the earth wire altogether.
- ELCB has some applications in various fields .
 - It is used in domestic, commercial, and industrial installations to provide protection against earth leakage currents.
 - It is used in outdoor and wet locations such as swimming pools, spas, fountains, and irrigation systems to prevent electric shock and electrocution.
 - It is used in medical and laboratory equipment to ensure the safety of patients and personnel.
 - It is used in power tools and appliances to prevent damage and fire hazards.

MCCB

MCCB stands for Moulded Case Circuit Breaker. It is a type of circuit breaker that protects the electrical circuit from overload, short circuit, and other faults. It is an advanced version of MCB (Miniature Circuit Breaker) that can handle higher currents and voltages. Some of the features of MCCB are:

- It has a moulded case or enclosure that contains the contacts, arc extinguishing mechanism, trip unit, and other components.
- It can be manually or automatically operated by a toggle or a motor.
- It has adjustable trip settings that can be changed according to the load and application requirements.
- It can provide protection against overcurrent, short circuit, ground fault, and residual current.
- It can be used for a wide range of voltages and frequencies, from 100 V to 1000 V and from 50 Hz to 60 Hz.
- It can interrupt currents up to 2500 A and withstand short circuit currents up to 100 kA.
- It can be classified into three types based on the trip unit: thermal magnetic, electronic, and magnetic only.
- Thermal magnetic MCCBs use a combination of thermal and magnetic elements to provide overload and short circuit protection. The thermal element responds to the average current and the magnetic element responds to the peak current.
- Electronic MCCBs use a microprocessor to measure the current and compare it with the preset values. They can provide more accurate and faster protection than thermal magnetic MCCBs. They can also offer additional features such as metering, communication, and diagnostics.
- Magnetic only MCCBs use only a magnetic element to provide short circuit protection. They do not offer overload protection and are usually used for motor protection.

ACB

- ACB stands for Air Circuit Breaker, which is an electrical device used to provide overcurrent and short-circuit protection for electric circuits over 800 Amps to 10kA .
- ACB operates at atmospheric pressure in air, where air-blast as an arc quenching medium .
- ACB is used in low voltage applications below 450V, such as distribution panels , and also in high voltage applications up to 15kV.
- ACB has three main components: a fixed contact, a moving contact and an arc runner .
- ACB has two types of operating mechanisms: spring operated and pneumatic .
- ACB has three types of arc extinction methods: plain air break, air chute and axial blast .
- ACB has several advantages, such as high speed of operation, low maintenance, high reliability, easy replacement and low noise .
- ACB has some disadvantages, such as high cost, large size, high air pressure requirement, and possibility of fire hazard .
- ACB has various applications, such as industrial plants, power stations, substations, switchboards, and motor control centers .

Types of Wires

Wires are conductors that carry electric current from a source to a load. They are usually made of metal, such as copper or aluminum, and have different sizes, shapes, and insulation materials. The type of wire used for a particular application depends on factors such as voltage, current, temperature, environment, and installation method.

Some of the common types of wires used in electrical engineering are:

- **Cleat wiring:** This is a temporary wiring system that uses porcelain or plastic cleats to hold wires on wooden boards or walls. The wires are usually insulated with rubber or cotton and are suitable for low voltage and low current applications. Cleat wiring is cheap and easy to install, but it is not durable, safe, or aesthetically pleasing .
- **Wooden casing and capping wiring:** This is a semi-permanent wiring system that uses wooden casing (a rectangular box) and wooden capping (a grooved strip) to enclose wires on walls or ceilings. The wires are usually insulated with rubber or PVC and are suitable for low and medium voltage applications. Wooden casing and capping wiring is more durable and safer than cleat wiring, but it is also more expensive and prone to fire and termite attacks .
- **CTS or TRS or PVC sheath wiring:** This is a permanent wiring system that uses tough rubber sheath (TRS) or polyvinyl chloride (PVC)

sheath to cover wires that are laid on wooden battens or clips. The wires are usually insulated with rubber or PVC and are suitable for low and medium voltage applications. CTS or TRS or PVC sheath wiring is more durable, safe, and economical than wooden casing and capping wiring, but it is also less flexible and more susceptible to moisture and chemical damage .

- **Lead sheathed or metal sheathed wiring:** This is a permanent wiring system that uses lead or metal (such as steel or aluminum) sheath to protect wires that are laid in metal conduits or pipes. The wires are usually insulated with rubber or PVC and are suitable for medium and high voltage applications. Lead sheathed or metal sheathed wiring is more durable, safe, and resistant to moisture, fire, and corrosion than CTS or TRS or PVC sheath wiring, but it is also more expensive and difficult to install .
- **Conduit wiring:** This is a permanent wiring system that uses rigid or flexible metal or plastic conduits or pipes to enclose wires that are laid inside walls, ceilings, or floors. The wires are usually insulated with rubber or PVC and are suitable for medium and high voltage applications. Conduit wiring is more durable, safe, and aesthetically pleasing than lead sheathed or metal sheathed wiring, but it is also more expensive and requires skilled labor .
- **Low-voltage wiring:** This is a special wiring system that uses wires with low voltage (usually 50 volts or less) and low current for applications such as landscape lighting, sprinkler systems, doorbells, speaker systems, and thermostats. The wires are usually made of copper or aluminum and have different sizes, colors, and insulation materials. Low-voltage wiring is more energy-efficient, safe, and flexible than high-voltage wiring, but it also requires transformers, switches, and connectors.
- **Hot, neutral, and grounding wires:** These are the three types of wires that are used in most household and commercial wiring systems. Hot wires are black or red and carry electric current from the source to the load. Neutral wires are white or light gray and carry electric current from the load back to the source. Grounding wires are bare copper or green and carry electric current in the event of a ground fault or short circuit. They help reduce the chance of severe electric shock or fire.

Cables and Bus-bars

- Cables and bus-bars are two methods of transporting electrical energy in distribution systems.
- Cables are insulated conductors that are usually made of copper or aluminum and can be run in trays or conduit. Cables can carry low, medium or high voltage currents and are flexible and easy to install. Cables can also be shielded to reduce electromagnetic interference.

- Bus-bars are metal bars that are usually made of copper or aluminum and are used to carry large amounts of current. Bus-bars are not insulated and are enclosed in a metal casing or duct. Bus-bars can carry low or medium voltage currents and are rigid and compact. Bus-bars can also be segmented to isolate faults and prevent short circuits.
- Some advantages of cables over bus-bars are:
 - Cables can be routed in any direction and can bend around obstacles.
 - Cables can be easily terminated and connected to different types of equipment.
 - Cables can be easily replaced or repaired if damaged.
- Some advantages of bus-bars over cables are:
 - Bus-bars can carry more current per unit cross-section than cables.
 - Bus-bars have lower resistance and impedance than cables, which reduces power losses and voltage drops.
 - Bus-bars have higher mechanical strength and thermal stability than cables, which reduces sagging and overheating.
 - Bus-bars have lower installation and maintenance costs than cables.

Fundamentals of earthing and lightning protection

- Earthing and lightning protection are important for everyone that uses electrical equipment and that includes the large majority of the world.
- Earthing is the process of connecting the metallic parts of an electrical system or appliance to the earth. This provides a low-resistance path for fault currents and protects the system and the users from electric shock.
- Lightning protection is the process of intercepting or diverting lightning and providing a certain path for conducting the surges safely to the ground by adequate down conductors to grounding electrodes. This helps prevent disastrous events like fires, injuries, and deaths.
- The basic principles of earthing and lightning protection are:
 - Providing an alternative path for the lightning current to flow to ground and ensure that:
 - * Lightning current flowing in the lightning protection system (LPS) does not induce dangerous currents in any parallel metallic systems near the LPS, and
 - * The potential difference between the LPS and the earth is minimized.
 - Providing a low-impedance path for the fault currents to flow to ground and ensure that:
 - * The fault current is quickly cleared by the protective devices, and
 - * The touch and step voltages are kept below the safe limits.
- The types and components of earthing and lightning protection systems are:
 - Earthing system: consists of earth electrodes, earth conductors, earth pits, and bonding conductors. The earth electrodes are buried in

the soil and provide a physical connection to the earth. The earth conductors connect the earth electrodes to the metallic parts of the electrical system or appliance. The earth pits are filled with conductive materials to improve the soil resistivity and moisture retention. The bonding conductors connect the metallic parts of the electrical system or appliance to each other and to the earth conductors.

- Lightning protection system: consists of air terminals, down conductors, and grounding electrodes. The air terminals are metal rods or wires that are installed at the highest points of a structure or equipment to intercept the lightning. The down conductors connect the air terminals to the grounding electrodes. The grounding electrodes are buried in the soil and provide a low-impedance path for the lightning current to dissipate into the earth.
- Surge protection devices: are devices that limit the voltage surges or transients that may occur due to lightning or switching operations. They are installed at the point of entry of the electrical supply or at the sensitive equipment. They can be classified into primary, secondary, and tertiary surge protection devices based on their location and function.

Types of Batteries

Batteries are devices that store electrical energy in the form of chemical energy and convert it into electrical current when needed. Batteries are widely used in electrical engineering for various applications, such as powering portable devices, electric vehicles, backup systems, etc.

There are mainly two categories of batteries: primary batteries and secondary batteries. A primary battery is a disposable kind of battery that cannot be recharged once used. A secondary battery is a rechargeable kind of battery that can be recharged again and again by applying an external voltage source.

There are also other types of batteries, such as reserve cells, fuel cells, and flow batteries, that have different characteristics and applications.

Some of the common types of batteries and their applications are:

- **Lead-acid batteries:** These are the oldest type of rechargeable batteries, invented in 1859. They consist of lead plates immersed in sulfuric acid solution. They have a low energy density, but a high power density and a low cost. They are used for automotive starter motors, emergency lighting, uninterruptible power supplies, etc.
- **Nickel-cadmium batteries (Ni-Cd):** These are rechargeable batteries that use nickel oxide hydroxide and metallic cadmium as electrodes. They have a high energy density, but suffer from memory effect and environmental issues. They are used for portable devices, cordless tools, medical equipment, etc.

- **Nickel-metal hydride batteries (Ni-MH):** These are rechargeable batteries that use nickel oxide hydroxide and a hydrogen-absorbing alloy as electrodes. They have a higher energy density and a lower memory effect than Ni-Cd batteries, but a lower power density and a higher self-discharge rate. They are used for consumer electronics, hybrid electric vehicles, etc.
- **Lithium-ion batteries (Li-ion):** These are rechargeable batteries that use lithium compounds as electrodes. They have a very high energy density, a high power density, and a low self-discharge rate. They are used for smartphones, laptops, electric vehicles, etc.
- **Alkaline batteries:** These are primary batteries that use zinc and manganese dioxide as electrodes. They have a high energy density, a long shelf life, and a low cost. They are used for flashlights, toys, remote controls, etc.
- **Zinc-carbon batteries:** These are primary batteries that use zinc and carbon as electrodes. They have a low energy density, a short shelf life, and a low cost. They are used for low-power devices, such as clocks, radios, etc.
- **Coin cell batteries:** These are small primary batteries that use various chemicals as electrodes, such as lithium, silver oxide, zinc-air, etc. They have a high energy density, a long shelf life, and a low cost. They are used for watches, calculators, hearing aids, etc.
- **Zinc-air batteries:** These are primary batteries that use zinc and oxygen from the air as electrodes. They have a high energy density, a low cost, and a long shelf life. They are used for hearing aids, electric vehicles, etc.
- **Sealed lead-acid batteries:** These are rechargeable batteries that use lead and sulfuric acid as electrodes, but are sealed in a plastic case. They have a low energy density, a high power density, and a low maintenance. They are used for backup power, security systems, etc.